

Least Squares

Overview

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- Extend the 1-D DFT to higher dimensions, and understand two-dimensional frequencies
- Get a geometric intuition of projection onto a subspace.
- Learn about least squares optimization and how they are solved using the Normal Equation.
- Learn about weighted least squares and how the normal equation is modified.
- How to use numerical libraries to solve linear least squares problems.
- How to generalize least squares to higher order polynomials as well as exponential and power fits.

Async

2D FFT

2D Discrete
Fourier
Transform

- a **vector space** is a space where you can **add** and **scale elements**
- **vectors** are just **elements** of a **vector space**
- two **identify orthogonal** elements, **inner product** is needed
- **2D arrays (tensors)** are **vectors** whose **inner product** is the **sum** of the **pairwise products** of the elements
- **create** and **array/vector**

$$v^{k_1, k_2} \in \mathbb{R}^{N_1 \times N_2}$$

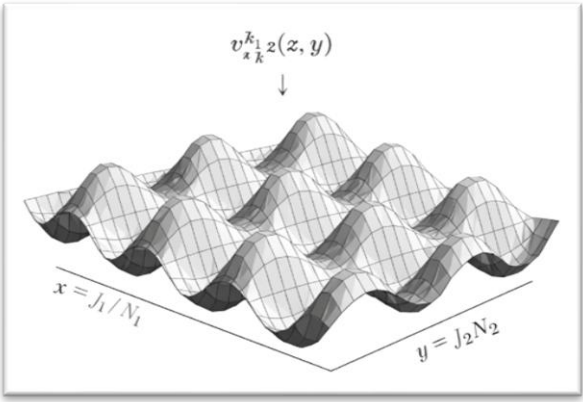
- with **two dimensions**

$$v_{j_1, j_2}^{k_1, k_2} = \exp \left(i \frac{2\pi j_1 k_1}{N_1} \right) \exp \left(i \frac{2\pi j_2 k_2}{N_2} \right)$$

Visualizing
Waves

- this **vector** is both **two-dimensional** as well as **complex**
- the **components** are **waves**

$$v_{j_1, j_2}^{k_1, k_2} = \sin \left(\frac{2\pi j_1 k_1}{N_1} \right) \sin \left(\frac{2\pi j_2 k_2}{N_2} \right)$$

	• normalized index coordinates $\left(\frac{j_1}{N_1}, \frac{j_2}{N_2}\right)$ • as a point in a unit square $x = \frac{j_1}{N_1}, \quad y = \frac{j_2}{N_2}$ • continuous domain representation $v^{k_1, k_2}(x, y) = \sin(2\pi k_1 x) \sin(2\pi k_2 y)$  ChatGPT 5.2.1
Orthogonal Vectors	• the vectors are orthogonal to each other $v_{j_1 j_2}^{k_1, k_2} = \exp\left(i \frac{2\pi j_1 k_1}{N_1}\right) \exp\left(i \frac{2\pi j_2 k_2}{N_2}\right)$ • orthogonality condition $v^{r_1, r_2} \cdot v^{s_1, s_2} = 0$ if $(r_1, r_2) \neq (s_1, s_2)$ • and the length of all the vectors are the same (vector norm) $ v^{k_1, k_2} ^2 = v^{k_1, k_2} \cdot v^{k_1, k_2} = N_1 N_2$ • the dot product is defined by listing all of the entries in a single list
Projection	
Solution	• the closest point p has the property that, $(x - p) \cdot v = 0$

	- for every vector $\mathbf{v}$ in $\mathbf{V}$, but that happens if, $(\mathbf{x} - \mathbf{p}) \cdot \mathbf{v}_i = 0$ - for every vector in the basis for $\mathbf{V}$, $\mathbf{v}_i^T \mathbf{x} = \mathbf{v}_i^T \mathbf{p}$ - since, $\mathbf{A} = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_n]$ - that means, $\mathbf{A}^T \mathbf{x} = \mathbf{A}^T \mathbf{p}$
QR Factorization	- to help with this, we use a very important factorization — QR. - given a matrix $\mathbf{A}$ with linearly independent columns, we can factor $\mathbf{A} = \mathbf{Q}\mathbf{R}$ - example: $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \\ 2 & 0 \\ 2 & 2 \end{bmatrix} = \mathbf{Q}\mathbf{R}$ $\mathbf{Q} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \mathbf{R} = \sqrt{3} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ $\mathbf{v}_1 = r_{11}\mathbf{q}_1$ $\mathbf{v}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$
Julia	<pre> julia> using LinearAlgebra julia> A = [2 1; 0 1; 2 0; 2 2]; julia> qrf = qr(A); julia> Matrix(qrf.Q) 4×2 Matrix{Float64}: -0.57735 2.29738e-17 -0.57735 -0.57735 -0.57735 0.57735 -0.57735 -0.57735 julia> R = qrf.R 2×2 Matrix{Float64}: -3.4641 -1.73205 0.0 -1.73205 </pre>

```
julia> Q = qrf.Q
4×2 Matrix{Float64}:
 2.0  1.0
 0.0  1.0
 2.0  0.0
 2.0  2.0

julia> Q'*Q
2×2 Matrix{Float64}:
 1.0      -2.01213e-17
-2.01213e-17  1.0
```

Use QR for Projection

- combine

$$A^T z = A^T x$$

$$A = QR$$

- get,

$$R^T Q^T z = R^T Q^T x$$

$$Q^T z = Q^T p$$

- **p** is in the **span** of **Q** columns

$$p = Qc$$

$$Q^T z = Q^T Qc = c$$

$$p = Qc = QQ^T z$$